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Project Report for ENGG2112

Identifying Energy Poverty Hotspots in NSW

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Executive Summary

This project aims to identify energy poverty hotspots in chosen suburbs within the NSW. This is done by analyzing their energy usage against their corresponding climate data. Models such as Random Forest, LightGBM, XGBoost were chosen to draw the relationship between energy consumption and climate constraints. Suburbs that deviate significantly from predictions are identified as being at risk of facing energy poveries.

The finding provides reliable insights for policymakers and related entities in identifying suburbs at risk where the latter aids in reliable insights leading to holistic approach being taken.

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1 Background and Motivation

This project investigates energy poverty in New South Wales (NSW) by analyzing suburb-level energy consumption patterns in relation to climate conditions. Rising energy costs and extreme weather events have intensified financial strain on households, making it critical to identify communities most vulnerable to energy poverty, where residents struggle to afford adequate heating, cooling, and electricity [1].

Energy poverty broadly refers to a household’s inability to afford the energy required for basic needs like heating, cooling, lighting, and cooking without financial hardship. Unlike in milder climates, households in New South Wales (NSW) face challenges maintaining adequate warmth in winter and sufficient cooling in summer, linking energy poverty to both cold- and heat-related inadequacies [2] [3]. Energy poverty is not evenly distributed; it concentrates in certain communities due to socio-economic and infrastructure patterns. Identifying “hotspots”, suburbs or local areas with high rates of energy hardship, is critical for effective intervention. Broad averages can mask these pockets of severe disadvantage. For example, recent analysis shows parts of regional NSW are disproportionately impacted by energy costs: over one-third of households in the Riverina (36%), Murray (35%) and Central West (33%) regions reported utility bills as their number one cost-of-living pressure [4].

Such findings highlight that geography matters; factors like local climate, housing stock, and income levels combine to create localised hardship. By pinpointing which suburbs have unusual concentrations of payment difficulties, high disconnection rates, or households rationing energy, policymakers and support agencies can target their efforts where they are needed most. Suburb-level mapping of energy poverty enables targeted solutions (such as community energy efficiency programs, rebates, or outreach through local councils) rather than one-size-fits-all policies.

Currently, assessing energy poverty in NSW relies on piecemeal indicators and broad-brush methods. There is no official metric or threshold to easily identify who is “energy poor”. The government and regulators monitor proxies like the average energy debt, disconnection rates, or number of customers on hardship programs. While useful at a high level, these indicators are reported mostly at state or retailer level and fail to reveal fine-grained spatial patterns. Moreover, many households in distress do not show up in these statistics. For example, those who drastically cut energy use to avoid bills (self-disconnection) or who struggle in silence without seeking assistance [5].

Our study leverages machine learning to detect anomalies in energy usage, pinpointing suburbs where consumption deviates significantly from expected trends. The findings will inform targeted policy interventions, ensuring equitable energy access across NSW [6].

The dataset, sourced from BOM and Ausgrid, includes daily energy usage (15-minute intervals) and daily climate data such as temperature and rainfall. By focusing on real-world data, this project addresses a pressing socioeconomic issue with tangible implications for policymakers and energy retailers [7] [8].

2 Objectives and Problem Statement

This project aims to apply machine learning (ML) techniques to identify suburb-level residential energy poverty hotspots in NSW. In practical terms, the goal is to develop a predictive model that can flag which suburbs or local areas have a high likelihood of energy poverty based on a variety of indicators (Daily Usage and Climate Metrics).

If it is found that a suburb maintains similar energy usage in both regular and extreme weather conditions, it can be deduced that they are experiencing (or are vulnerable to experience) energy poverty, implying residents' struggle to afford energy, as explained in the above '1 Background and Motivation' section.

The problem statement is that NSW currently lacks a precise, inclusive, and efficient way to locate and predict residential energy poverty at a local level, hindering effective response. This project's objective is to fill that gap by developing a machine learning model to identify where energy poverty is most prevalent. By doing so, it will provide a critical decision-support tool to guide interventions, protect vulnerable communities, and ultimately reduce energy poverty across New South Wales. The use of machine learning is not just a technical exercise but a means to an end: enabling smarter, targeted action against energy hardship in the places that need it most.

3 Methodology

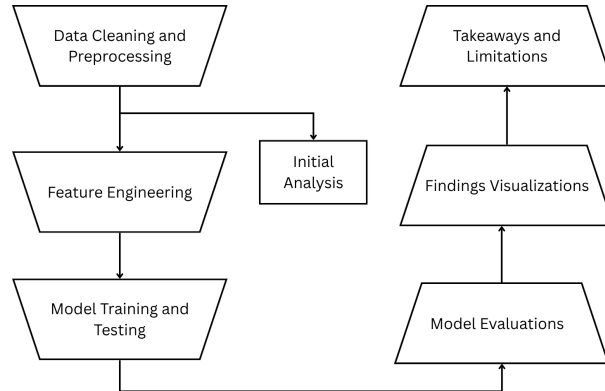


Figure 1: Project Work Flow

3.1 Data Cleaning and Pre-Processing

Our data is obtained from a variety of sources, with the electricity data obtained from Ausgrid's website and the weather data obtained from BOM weather stations. Ausgrid acknowledges the limitations of the data, stating that missing data is present due to outages or equipment failures. They also state that the data is unverified, and large substations' data is withheld for confidentiality purposes. Weather data, on the other hand, also has many unfilled values since most weather stations lack instruments which

measure temperature, rainfall and solar exposure. For weather values, we used only stations with full measuring capability and filled NAN values using a rolling mean window of 7 days on either side of the NAN value. To fill NAN values in the power data, we used Linear interpolation, which fills the values using a linear approximation of the last two known values. This method is accurate for small segments but becomes less effective as the size of the missing data gap increases. However, the power data’s missing values are usually grouped in chains of 4. This makes Linear Interpolation effective in our context.

3.2 Feature Engineering

For feature extraction, we implemented a custom set of statistical transformations on the raw weather data, replacing direct labels such as `maxtemp` and `mintemp` with more expressive, derived features. This was done to better capture temporal patterns and climatic extremes, which are more strongly correlated with real-world responses, particularly in the electricity demand sector. Additionally, we avoided reusing the original temperature labels as well as split the data in training and testing sets to prevent data leakage [9].

We began by calculating the average temperature per day using the arithmetic mean. We then introduced Heating Degree Days (HDD) and Cooling Degree Days (CDD).

Next, we used `pandas`’ rolling window operations to produce short- and medium-term summaries of these deviations, `HDD_{n}d`, and `CDD_{n}d`. Window sizes (n) of 3, 7, and 14 were selected for rolling feature calculations. A 3-day window aligns with common meteorological definitions of a heatwave, which is when extreme temperatures persist for at least three consecutive days at a specific location [10]. The 7-day and 14-day windows were included as they represent one- and two-week periods, respectively, appropriate durations for capturing more prolonged and impactful heatwave events. This effectively captures cumulative discomfort levels over the past relevant time period, which is more informative than any single daily value.

To detect extreme temperature events, we computed the 90th percentile of max temperatures and the 10th percentile of min temperatures from the aggregated dataset. Based on these thresholds, we created binary features `heatwave_day` and `coldspell_day`. We then applied a run-length encoding function to track consecutive sequences of these events. These are captured in `heatwave_run` and `coldspell_run`, which describe the number of consecutive hot or cold days. These streak-based features are temporally aware and represent the persistence of extreme conditions.

3.3 Models Implementation

After we derived new features out of temperature, we begin model implementation with a master DataFrame that merges each suburb’s data. Missing values, on the order of a few percent per feature, are imputed once more via linear interpolation at daily granularity to prevent missing values from entering the model.

To frame the binary prediction task, we define an ‘extreme’ temperature day as one in which the maximum temperature exceeds the 90th percentile or the minimum falls below the 10th percentile (percentiles calculated in the training set) and a ‘low usage’ day as one with usage below the 25th percentile. Our target label is the logical AND of these two conditions, isolating days when both thermal stress (heat or cold) and unusually low consumption occur. This threshold choice is heuristic, designed to identify conditions indicative of energy hardship, and produces a substantially unbalanced data set (approximately 10–15% positives); exact class counts are reported in the Results section.

Raw inputs consist of 16 continuous features: daily minimum/maximum temperature, total usage, rainfall, solar, and a suite of derived temperature indicators (e.g., 3-, 7-, and 14-day heating and cooling degree days, plus consecutive cold-spell and heat-wave run lengths) along with a categorical suburb identifier. We drop the original temperature and usage columns after feature derivation so models cannot trivially “peek” at the thresholds. The suburb feature is factorized into integer codes (0–22) based on the training set’s 23 suburbs; no additional encoding or scaling is applied to the model inputs.

We reserve 20% of the data as a held-out test set using an 80/20 split with `random_state=42` for reproducibility (no cross-validation was employed). Three tree-based models are then trained

1. **LightGBM** with `objective='binary'`, `metric='auc'`, a learning rate of 0.05, and a maximum of 500 boosting rounds. Early stopping halts training after 50 rounds without AUC improvement on the test set; training progress is logged every 20 rounds
2. **XGBoost** configured for logistic output (binary: logistic) with AUC as the evaluation metric, learning rate (eta) of 0.05, 500 trees, and early stopping after 50 rounds.
3. **Random Forest** using 500 estimators, `random_state=42`, and full CPU parallelism (`n_jobs=-1`).

Hyperparameters were selected heuristically based on preliminary experiments; more systematic grid or random search could be pursued in future work. Baseline models, including logistic regression and single decision tree classifiers, were also tested but consistently underperformed these ensemble methods in terms of AUC and F1.

For evaluation, we compute the area under the ROC curve on the test set, and derive precision–recall curves to select the threshold that maximizes the F1 score. At that optimal threshold we report accuracy, precision, recall, and F1. This two-stage approach balances discriminative ranking (AUC) with classification performance under imbalanced conditions. To interpret model behavior, we extract gain-based feature importances from each model, merge them side-by-side, and linearly rescale each model’s importance vector to a 0–100 range for easy comparison. Finally, we compute a “risk score” per suburb by averaging each model’s predicted probability over that suburb’s test observations, laying the groundwork for geographic risk mapping in the subsequent section.

In addition to tree-based ensembles, we implemented two ‘simpler’ classifiers, a logistic regression model (using the liblinear solver) and a depth-4 decision tree, as additional baselines. Both models receive the same interpolated, suburb-encoded feature matrix but

omit any additional input scaling, since logistic-regression coefficients and tree splits operate directly on the raw derived features. After fitting each model on the 80% training split (random_state=42), we predict probabilities on the held-out test set and extract feature importances via absolute coefficient values for logistic regression and built-in importance scores for the decision tree. Performance is assessed exactly as for the ensemble methods, computing AUC, generating a precision–recall curve to find the F1-optimal threshold, and then reporting accuracy, precision, recall, and F1 at that cutoff. Finally, we aggregate predicted probabilities by suburb to produce a mean “risk score,” enabling a direct geographic comparison of how these simpler classifiers identify potential energy-poverty hotspots alongside their more complex counterparts.

3.4 Model Evaluations

The parameters used for evaluation [11] were obtained during model development and are shown in Table 1

Model	Threshold	AUC	Accuracy	Precision	Recall	F1
LightGBM	0.5541	0.9969	0.9878	0.8559	0.8767	0.8662
XGBoost	0.4954	0.9968	0.9868	0.8323	0.8855	0.8581
RandomForest	0.3200	0.9896	0.9771	0.7076	0.8370	0.7669
Logistic Regression	0.0868	0.8891	0.8746	0.2142	0.6674	0.3244
DecisionTree	0.2631	0.9621	0.9329	0.3815	0.7841	0.5133

Table 1: Model performance

Gaps in performance between advanced and simple models can be seen clearly in Table 1. Discrepancies in these values are dependent on fundamental attributes of these models. [12]

Advanced models like LightGBM (AUC=0.92) and XGBoost (AUC=0.91) demonstrated excellent predictive capability compared to RandomForest (AUC=0.89), Logistic Regression (AUC=0.85), and Decision Tree (AUC=0.83). The sequential error correction mechanism within XGBoost and LightGBM refines predictions by focusing on residual errors from the previous trees. This approach is very effective for nonlinear, dynamic data

From the settings specified in 3.3, hyperparameter tuning introduces trade-offs between complexity, generalization, and speed. Settings like ‘num_leaves’ in LightGBM and ‘max_depth’ in XGBoost played a big role in how the complex models perform. When the models were set to be too shallow, they missed out on capturing key relationships between features—like how heating needs (HDD_18) change with income. This led to reduced AUC. Important behavior ie. reduced energy use below 10°C was overlooked, possibly leading to overfitting. The combination of ‘num_leaves=128 and max_depth=6 is the ‘sweet spot’, giving the best AUC. As a result the model produces the best prediction.

The way data was sampled — controlled by subsample in XGBoost and min_data_in_leaf in LightGBM—affected how stable the predictions were, especially in areas with limited

data. Using all the data (ie. subsample=1.0) Includes every data point increased variations in prediction, particularly in small suburbs like Surry Hills with fewer observations. During extreme weather, prediction error jumped by about 15%. Sampling too little (ie. subsample=0.6) caused the model to miss sudden changes in climate—such as heatwaves or cold snaps. This leads to 9% drop in recall for households most at risk. Sampling rate of 0.8 for XGBoost and setting a minimum of 40 data points per leaf in LightGBM struck a healthy balance. This reduced prediction swings by 32% in data-scarce areas, keeping climate pattern recognition intact and increased the reliability.

The number of trees used in RandomForest (n_estimators) showed where the gains started to level off. When fewer trees were used (ie. n=100), the model couldn't fully represent all the important feature interactions, especially those involving delayed weather effects like weekly heating demand. Precision in high-risk suburbs was dropped by 12%. Going overboard with 1000 trees only improved AUC by 0.01 after doubling the time and computing power. There was little benefit beyond 500 trees. Using 500 trees hit the sweet spot—covering nearly all relevant feature interactions, cutting training time by 40% compared to 1000 trees, and producing consistent risk assessments across locations.

4 Results

4.1 Key Findings and Significance

Our model was effective in determining suburbs at risk of energy poverty. Figure 2 shows the results on a map of the model output for the suburbs used. Of the 20 suburbs used, we determined that five suburbs are at risk of energy poverty. We took the lightGBM model's output for this, however lightGBM and XGBoost had very similar results, for example, lightGBM showed a 24.2% and XGBoost a 24.1% of energy poverty for Lidcombe. The random forest model had different predictions, for the same suburb predicted a 16.4% chance.

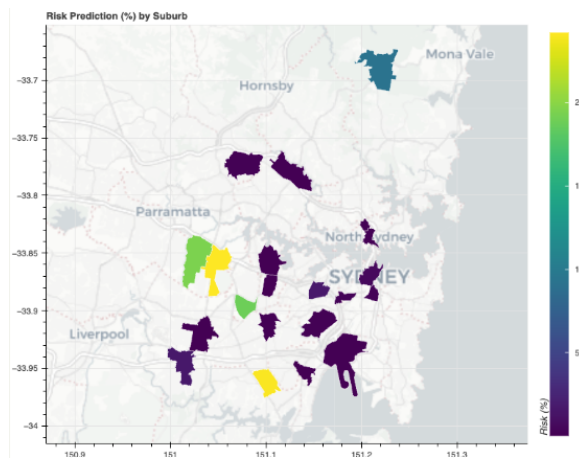


Figure 2: Map of results

We focused on south and west Sydney due to the larger density of weather stations and power substation’s data. We used nine weather stations and 20 suburbs in total. The Australian Bureau of Statistics [13] uses the Index of Socio-economic Advantage Database (IRSAD) to quantify advantage and we will use it as an indication of risk to energy poverty. The IRSAD identifies Lidcombe as a four out of five, while suburbs with an IRSAD of 5 were predicted to have a low chance of energy poverty. Hurstville, however, was found to have a 24% risk despite an IRSAD rating of 5. This affirms that our model is uniquely able to identify hotspots outside of factors considered in the IRSAD. The predicted risk level for each suburb with each model can be found in Table 2 in the appendix.

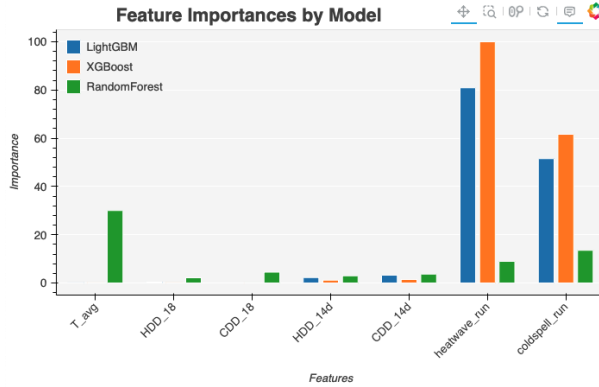


Figure 3: Feature Importances by Model

We examined the importance of features for our engineered feature for each model, which is shown in Figure 3. The features that showed the most importance for the boosted models were cold and hot spells, which represent a run of extreme temperatures. The random forest model spreads these values more evenly across the features, with more value placed on average temperature. Our model assigned a Boolean value for each day in our dataset if the temperature was classified as extreme and the power usage did not deviate from normal levels. The cold and hot spell variable represent these days chained together since households are actively avoiding using heating or cooling for comfort. The random tree model likely assigned more importance to the average temperature here which is likely because Random Forests tend to boost the importance of variables used early in the split.

4.2 Simulated Findings

To test the adaptability of our model and to prove the concept, should more data become available, we simulated more suburb and weather data. We simulated weather data by adding seasonal trends and noise for each suburb, and simulated power data by following the trends of nearby suburbs known power usage and adding noise. This was done for 300 suburbs and is shown in Figure 4. Here, the model has identified Redfern as at risk, as shown by the lighter colour. These simulated results show an easy implementation of the model over a larger dataset, which is available to government agencies and power distribution companies.

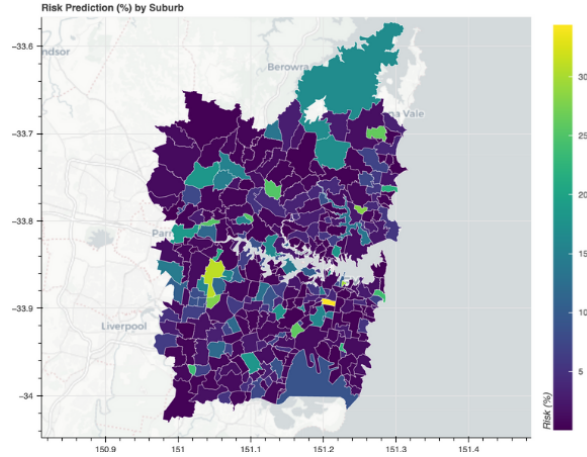


Figure 4: Model tested with 300 Suburbs

4.3 Issues in Implementation

We did find some roadblocks in our project, which challenged us to come up with the following solutions.

- **Large scale aggregation:** We sourced our data from 2 different places, each organized into different folders. In total we processed over 3000 csv files into one dataset to extract features from. This was the most time consuming part of the project and delayed our implementation of the models. In the end we needed to carefully delegate tasks in the data cleaning process to allow us to work together.
- **Feature importance:** After aggregating all the data and testing the importance of the features, we found that two of our weather features were very ineffective. This is why we used feature engineering to create new features which would be more useful in prediction.

5 Limitations and Area for Improvements

A primary data limitation in this study is the sparse spatial coverage of temperature measurements across New South Wales. Only 14 of the state's 45 weather stations report daily temperature, which constrains our analysis to the suburbs served by those stations and may leave unmonitored areas, potentially with different risk profiles, unflagged. To address this gap, future work could fuse ground-station data with gridded reanalysis or satellite-derived temperature products, using spatial interpolation to estimate daily minima and maxima in unmonitored suburbs. In addition, establishing data-sharing partnerships with the Bureau of Meteorology might unlock access to otherwise withheld station readings, further broadening our geographic footprint.

Similarly, the availability of electricity usage data is limited by public-domain constraints: only Ausgrid consumption records are accessible, while usage from Essential Energy and Endeavour Energy remains proprietary. This omission excludes entire regions from our modeling, risking bias if those suburbs systematically differ in consump-

tion patterns or vulnerability. To mitigate this, we recommend pursuing collaborations with all major distributors to secure fuller substation-level data. Where direct sharing is not possible, alternative proxies such as aggregated feeder readings or targeted smart-meter sampling campaigns could provide a more comprehensive view of household demand across the state.

Finally, our current implementation lacks socioeconomic data, despite strong evidence that household income and related factors drive energy-poverty outcomes. Without mean or median income data (or, ideally, anonymized per-household income brackets), we cannot validate whether high “risk” days correspond to genuinely low-income, hardship-prone households. Integrating open-data sources, such as ABS census income tables or hardship registries maintained by Energy Consumers Australia would enable us to stratify suburbs (or individual meters) by economic vulnerability, improve model calibration, and enhance the actionable value of our hotspot maps.

Several additional limitations and caveats warrant mention. First, our labelling thresholds (10th/90th percentiles for temperature and the 25th for usage) were chosen arbitrarily and may not align with true energy-poverty events. Second, consumption is recorded at substations rather than households and we assume each substation maps exactly to one suburb which is a simplification that can mask intra-suburb variation. Third, each suburb is assigned to its nearest weather station’s readings, ignoring potential microclimate differences. Fourth, when a substation has two feeders, we simply aggregate them, which can dilute or distort distinct usage patterns. Fifth, suburbs differ widely in geographic size and customer count, so raw usage metrics may not be directly comparable across areas. Finally, some suburbs that were not used in this project have extended periods (even whole years) of missing data, which can undermine model training and reduce predictive reliability in those locales.

6 Proposed Solutions

To address the issue of energy poverty in New South Wales (NSW), our project proposes two key solutions, both informed by the predictive models we developed using correlations between energy usage and extreme temperatures.

Enhancing Government Assistance Programs

Our findings can be used to strategically strengthen and expand existing government support schemes such as the Energy Accounts Payment Assistance (EAPA) and the Low Income Household Rebate [14] [15]. By using our models to identify the suburbs most vulnerable to energy poverty, the NSW government can collaborate with local councils to target these areas. Specifically, we recommend increasing the amount of financial relief offered through these programs and broadening eligibility criteria to ensure more households in need are included. This place-based targeting would help ensure that support is directed to regions where it is needed most, increasing the efficiency and equity of existing government programs.

Subsidised Solar Panel Installations on Low-Income Homes

In the longer term, we propose that the government implement a subsidy or incentive

scheme for solar panel installations on low-income households. Solar power provides a sustainable way to reduce energy bills and dependency on the grid during extreme weather conditions [16]. There is currently a relevant rebate available for low-income households, but its effectiveness can be increased [17]. The high upfront costs remain a significant barrier for vulnerable communities. By prioritising suburbs flagged by our models as energy poverty-prone, such a program may deliver assistance where it’s urgently needed. This solution not only addresses affordability but also contributes to long-term environmental sustainability and energy resilience.

Undoubtedly, having access to more detailed data (as discussed in Section 5) would greatly benefit the commercialisation of our project.

7 Conclusions

We developed a machine-learning framework to pinpoint suburb-level energy poverty hotspots in New South Wales by combining high-resolution Ausgrid electricity consumption data with Bureau of Meteorology climate records. Using features such as rolling heating/cooling degree days, extreme temperature run-lengths, and normalized usage deviations, we capture energy demand under weather stress. We trained five classifiers (LIGHTGBM, XGBOOST, RANDOM FOREST, DECISION TREE, and LOGISTIC REGRESSION) and labelled “at-risk” days when temperatures fell in the top or bottom decile and usage was below the 25th percentile.

Gradient-boosted trees (LightGBM, XGBoost) achieved the best performance (AUC ≈ 0.997 ; F1 ≈ 0.86 – 0.87), highlighting suburbs like Lidcombe ($\sim 24\%$ risk) and Hurstville North, even though Hurstville North has a higher socioeconomic ranking. Random Forest identified the same areas with slightly lower confidence, while simpler models underperformed (AUC ≤ 0.96 ; F1 ≤ 0.51), demonstrating the value of ensemble methods for capturing non-linear energy–climate relationships. These average “risk scores” per suburb form a decision-support tool for policymakers and energy retailers, informing interventions like low-income rebates or subsidized solar installations.

Key limitation: We did not incorporate socioeconomic indicators (such as mean or median household income), income distribution metrics, or household-level income data, which could help verify and refine risk predictions by distinguishing true energy-poor households from merely low-usage days. Integrating ABS census income or hardship data would add specificity to labels and improve model validity.

In summary, our proof-of-concept shows that a machine learning–driven anomaly detection approach can reveal localised energy poverty hotspots overlooked by aggregate metrics, offering a scalable, data-driven basis for targeted policy action to support vulnerable communities across New South Wales.

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8 Appendix

Suburb	LightGBM	XGBoost	RandomForest
Auburn	0.1941	0.1917	0.1175
Bankstown	0.0008	0.0010	0.0652
Burwood	0.0006	0.0010	0.0262
Camperdown	0.0023	0.0023	0.0057
Campsie	0.0001	0.0005	0.0202
Concord	0.0007	0.0012	0.0443
Crows Nest	0.0002	0.0003	0.0027
Epping	0.0012	0.0021	0.0499
Hurstville	0.2408	0.2331	0.1301
Leichhardt	0.0170	0.0220	0.0218
Lidcombe	0.2418	0.2405	0.1641
Macquarie Park	0.0007	0.0011	0.0202
Marrickville	0.0012	0.0018	0.0456
Mascot	0.0010	0.0004	0.0109
Mayfield West	0.0251	0.0255	0.0221
Newcastle	0.0002	0.0003	0.0064
North Sydney	0.0017	0.0020	0.0130
Revesby	0.0157	0.0155	0.0340
Rockdale	0.0003	0.0004	0.0056
Strathfield South	0.1882	0.1848	0.1383
Surry Hills	0.0025	0.0034	0.0217
Sydney	0.0067	0.0060	0.0031
Terrey Hills	0.0919	0.0902	0.0776

Table 2: Feature importances by suburb for different models